

FINAL EVALUATION

PART 1: Analysis of a small data-set | BESE-300

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1 Data-set used in the analysis

The present work analyses the fictional data-set ‘DATA_SET_REFERENCE_EVAL.csv’, which presents 400 measurements of HDL and LDL in patients across two planets (Earth and Venus) from two populations (tall and small), as seen in Table 1. No further information was given about the dataset due to it being fictional; nonetheless, in a real one, the source must be trustworthy with traceable methods.

Table 1: Fictional Data-Set. First 3 entries

	HDL	LDL	Size	Planet
Patient1	222.0001	114.1767	Tall	Venus
Patient2	150.3521	161.5871	Tall	Earth
Patient3	188.2625	114.2765	Tall	Venus

1.1 Overview

- **Total Records:** 400 patients.
- **Locations:** Distributed nearly evenly across two planets (204 on Venus, 196 on Earth).
- **Demographics:** Exactly split by size (200 Tall, 200 Small).
- **Columns:**
 - **Patient:** Unique identifier (Patient1 to Patient400).
 - **HDL (High-Density Lipoprotein levels):** Numerical variable (mg/dL).
 - **LDL (Low-Density Lipoprotein levels) :** Numerical variable (mg/dL).
 - **Size:** Categorical variable (Tall or Small).
 - **Planet:** Categorical variable (Venus or Earth).

Both columns, HDL and LDL, contain negative values (something impossible) and one NA value each, telling us that the data must be cleaned. Apart from that, all the other columns seem to be in order, without typos or strange categories; therefore, most of the attention will be centered on the columns HDL and LDL.

2 Curation of the data

2.1 Negative and NA values

It was decided to remove each patient who has a negative or NA value. For this particular data-set, both negative and NA

values for HDL and LDL corresponded to only 2 patients as seen in Table 2.

Table 2: Negative and NA Values

	HDL	LDL	Size	Planet
Patient8	NA	-30	Tall	Earth
Patient11	-20	NA	Tall	Earth

2.2 Outliers

Outliers were defined using a threshold of three standard deviations from the mean. For each numerical column in the dataset, we calculated the upper limit (Equation 1) and lower limit (Equation 2) as follows:

$$\text{Upper Limit} = \text{mean} + 3 \times \text{SD} \quad (1)$$

$$\text{Lower Limit} = \text{mean} - 3 \times \text{SD} \quad (2)$$

However, after applying these filters, no values were excluded, resulting in the same 398 observations remaining from the ‘Negative and NA values’ step.

3 Data exploration

3.1 Normalization and balance

Although measurement units were not provided, HDL and LDL are assumed to be in milligrams per deciliter (mg/dL) based on the magnitude of the numbers and a comparison with average population values reported by Tsao et al. (2023) (~111 mg/dL and ~53 mg/dL, respectively). All values in the dataset are biologically possible, though some are exceptionally rare. While literature ranges are often used to filter extreme values or check data quality, this strategy was not applied here given the fictional nature of the dataset.

The HDL and LDL distributions were explored using histograms and Q-Q plots; only the Q-Q plots are shown here for brevity (Figure 1 & Figure 2). The visualizations suggest that both variables follow a normal distribution (later confirmed by a statistical test); therefore, there was no need for data transformation. Finally, a cross-tabulation (Table 3) confirmed that the comparison groups are balanced.

3.2 Comparing Categorical and Continuous Data

The numerical variables (HDL and LDL) were grouped by Size and Planet and visualized using boxplots. There appeared to be no significant difference in LDL or HDL when grouping by planet; however, analyzing Figure 3 and Figure 4 suggests a notable trend when grouping by size, indicating a potential difference in HDL and LDL levels between tall and small individuals.

Table 3

Planet	Size	n	Percent
Earth	Small	99	24.9
Earth	Tall	95	23.9
Venus	Small	101	25.4
Venus	Tall	103	25.9

Sample sizes by Planet and Size group

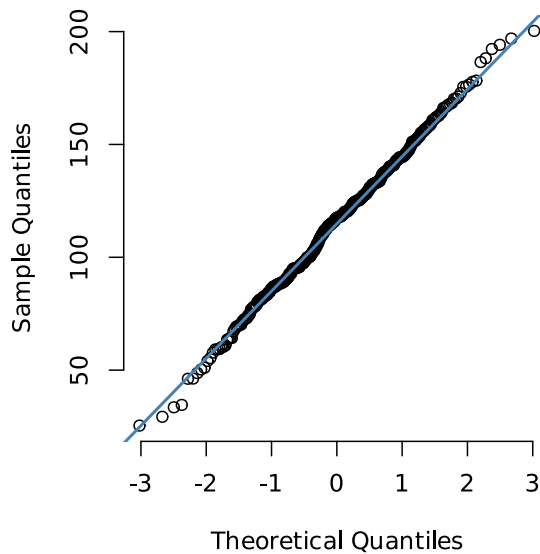


Figure 1: Q-Q Plot of LDL levels

3.3 Comparing Continuous Data

To explore the correlation between the continuous variables, LDL and HDL, a scatterplot was created (Figure 5), using color to indicate Size (Tall or Small) and shape for Planet (Earth and Venus). The plot reveals a negative trend: regardless of planet, small patients tend to have lower LDL and higher HDL levels, while the inverse is true for tall individuals.

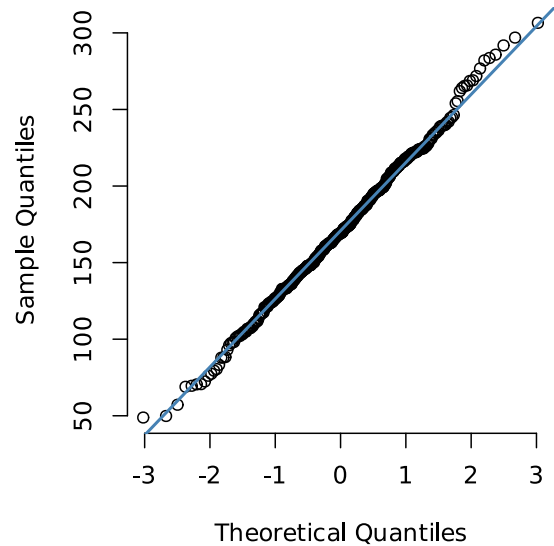


Figure 2: Q-Q Plot of HDL levels

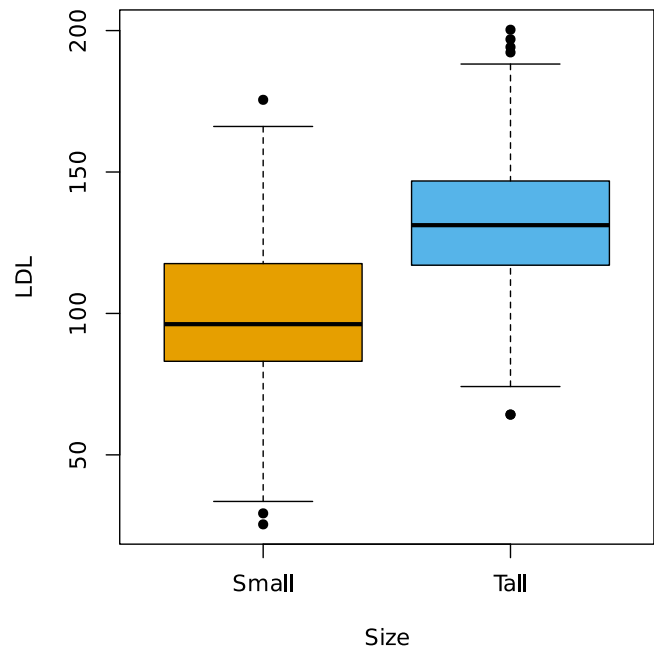


Figure 3: LDL levels by Size group

4 Statistics used to validate associations

4.1 Normality Analysis

Shapiro-Wilk was used to analyze normality in this set due to its size (398 samples < 5000) (Royston 1982). All the results were compared and validated with histograms and Q-Q plots (only the latter are shown in this report for brevity).

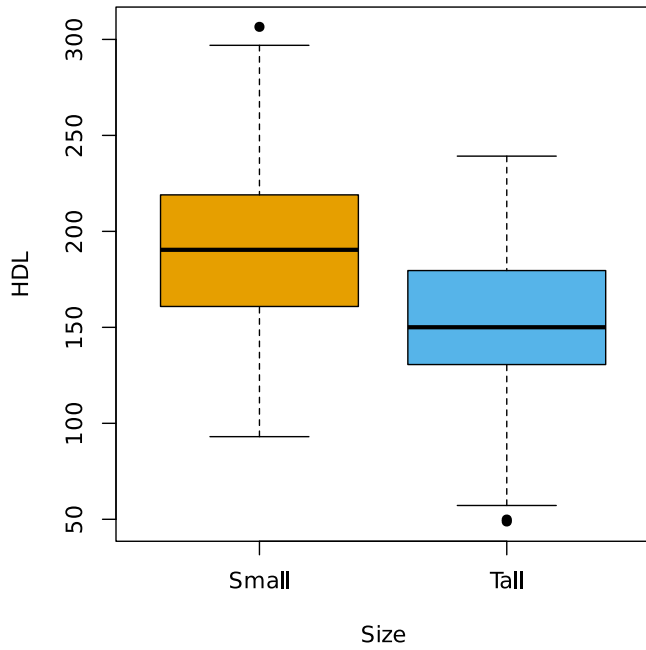


Figure 4: HDL levels by Size group

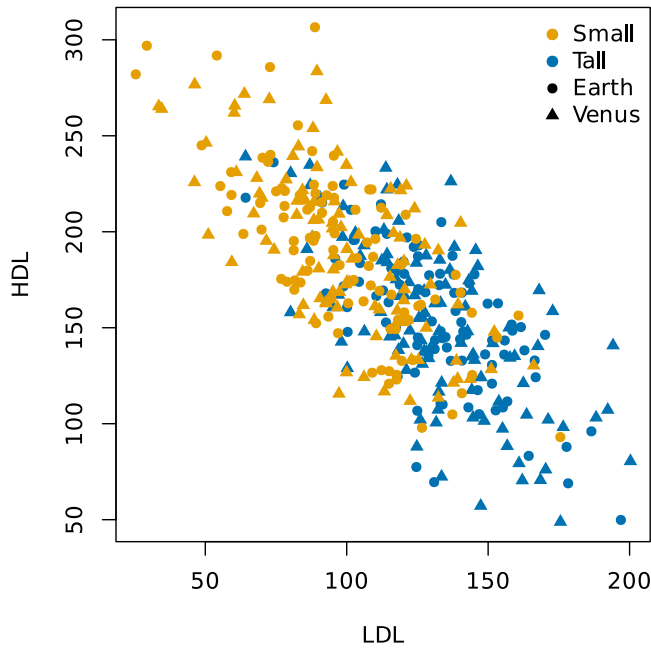


Figure 5: HDL vs LDL by Size (color) and Planet (shape)

4.2 Statistical Tests

Due to the samples number that causes a low loss in statistical power, to evaluate difference between comparative groups Welch’s two-sample t-test was performed over the traditional Student’s t-test.

4.3 One Analysis of Two Continuous Variables

Figure 5 suggested a potential negative correlation between HDL and LDL. To formally evaluate this relationship, the normality of both variables was first assessed using the Shapiro-Wilk test (Table 4). The results for both variables yielded p –

values > 0.05 , failing to reject the null hypothesis. Because the assumption of normality was met, Pearson’s parametric correlation test was selected to analyze the data.

The test revealed a strong negative correlation between the two variables (Pearson’s $r = -0.745$, $p = 1.24e - 71$), and the small p-value confirms this is very unlikely to be due to chance. This inverse relationship doesn’t imply causation, but is an expected result that is supported by existing literature (Burgess et al. 2014).

Table 4: Shapiro-Wilk normality test for HDL and LDL

Variable	Statistic	p-value	Conclusion
HDL	0.997	7.25e-01	Fail to reject H0 (normal)
LDL	0.997	8.09e-01	Fail to reject H0 (normal)

4.4 One Analysis of One Continuous Variable vs One Categorical Variable

As discussed before, an initial visual inspection of Figure 3 and Figure 4 suggested an apparent difference in HDL and LDL levels between tall and small individuals, regardless of planet. To statistically corroborate these observations, the data was grouped first by Size, and subsequently by Planet.

To determine the appropriate statistical comparison, each subset was evaluated for normality using the Shapiro-Wilk test. All groups yielded a $p - value > 0.05$, thereby failing to reject the null hypothesis: data follows a normal distribution (**?@tbl-normality-groups**). Consequently, parametric independent Welch T-tests were chosen over the non-parametric Kruskal-Wallis test for group comparisons, as detailed in Table 5.

As suspected, the T-test results comparing the mean differences of HDL and LDL grouped by Size (tall vs. small) yielded highly significant p-values ($p < 0.001$), firmly rejecting the null hypothesis of equal means. Conversely, this significant difference is entirely absent when grouping HDL or LDL by Planet ($p > 0.05$). Therefore, we can conclude that physical size, rather than planetary origin, is the primary driver of the differences observed among these groups.

Table 5: Comparison groups

Comparison	Test	Statistic	p-value	Conclusion
Tall vs Small (HDL)	T-test	-9.053	6.46e-18	Sig.
Tall vs Small (LDL)	T-test	12.899	5.12e-32	Sig.
Earth vs Venus (HDL)	T-test	0.212	8.32e-01	N.S.
Earth vs Venus (LDL)	T-test	-0.270	7.87e-01	N.S.

References

Burgess, Stephen, Daniel F Freitag, Hassan Khan, Denis N Gorman, and Simon G Thompson. 2014. “Using Multivariable Mendelian Randomization to Disentangle the Causal Effects of Lipid Fractions.” *PloS One* 9 (10): e108891.

Royston, J Patrick. 1982. "An Extension of Shapiro and Wilk's w Test for Normality to Large Samples." *Journal of the Royal Statistical Society: Series C (Applied Statistics)* 31 (2): 115–24.

Tsao, Connie W, Aaron W Aday, Zaid I Almarzooq, et al. 2023. "Heart Disease and Stroke Statistics—2023 Update: A Report from the American Heart Association." *Circulation* 147 (8): e93–621.